

Performance Analysis of Melanoma Early Detection using Skin Lession Classification System

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Abstract - Melanoma has been proved to be very tedious and statistical analysis which provides the majority of deaths occurs from skin cancer. The earlier detection and treatment is the best way of survival from melanoma. Clinical diagnosis of melanoma is extremely difficult due to its irregularity in edges and its shape. This paper proposes a novel scheme for early detection of melanoma using Multiclass support vector machine (MSVM). There are five different skin lesions which are grouped as Solar Keratosis or actinic keratosis, Basal Cell Cancer, Nevocytic nevus, Squamous Cell Cancer, Seborrhoeic Verruca. The proposed system uses an automatic procedure, where the queried images are grouped and matched with higher probability type to classify the type of melanoma. The multi class [6][7] support vector machine is a powerful tool for solving classification problem. The algorithm is based on learning of each stage with some training sample. Here, the color and texture features such as gradient, contrast, edges are extracted. The proposed system contains an image database which has the all five types of melanoma for testing and classification purposes. From the result of simulation, the accuracy of the proposed support vector machine scheme has comparatively high among all five types.

Keywords - MSVM, GLCM, RGB, OAA

I. INTRODUCTION

Melanoma is generally known as malignant melanoma, where it is a type of cancer that occurs in the pigment containing cells in epidermis layer called as melanocytes. Melanoma is curable by treatment, if the detection is at earlier stages. Mostly melanomas occurs in eye, intestines and mouth, where for women's the mostly occur in legs and mostly back for men's. The melanoma

can be easily identified by the change in size of moles, irregularity in edges, color change and itchiness of skin. The rate of survival of patients for last five year analysis shows that only 15% of survival of the patients has undergone the treatment at the chronic stages. The melanoma is steadily increasing skin cancer [9] in last 25 years, for those we are in a need of an urgent procedure for earlier detection systems. The earlier method was noninvasive imaging tools that are used by dermatologists which are called as epiluminescence microscopy. In order to use this tool, a specially trained dermatologists can able to analyze using a set of complex visual patterns such as lines, pigmented networks, blue veil, white veil, marks and blobs.

The result analysis is based on the presence and absence of these complex visual patterns diagnosis and can be derived by a dermoscopic [2] algorithm. A highly trained dermatologist is needed in noninvasive technique for diagnosing the dermoscopic image. Even if with the help of highly trained dermatologists the accuracy of diagnosing has been less and also the time required for diagnosing will be longer. In other words, in order to achieve diagnosing of melanoma, the clinicians should undergo a highly extensive training. So that, the clinicians can be able to analyze the features and patterns from the dermoscopic images.

Untrained clinicians are not allowed for diagnose because of trained professions sometimes misjudging of diagnoses will occur. This paper proposes an automated system for analyzing melanoma type classification and early detection. In this paper, it is possible to concentrates on five types of melanoma skin cancers [8] and also finding the region at which cancer cells are present are also described for therapeutic purposes. The proposed system is a automated based classification where the image is

classified by matching. The proposed system contains image acquisition, color space conversion, feature extraction, classifier algorithm and segmentation.

1.1 Related Works

There are several classification methods have been introduced and all those methods had used an different algorithms for classification and also the accuracy and load of those existing methods vary with each other. The proposed methods are used to extract the features from the image and to classify the cancer from the given predefined sets.

II. SYSTEM MODEL

In this paper, it is possible to propose an automated diagnosis system which is used to extract the color features and texture descriptors. Multiclass support vector machine classifier is used to diagnose and classify the skin lesions [3][5] in a feasible way.

2.1 System Model of Proposed Method

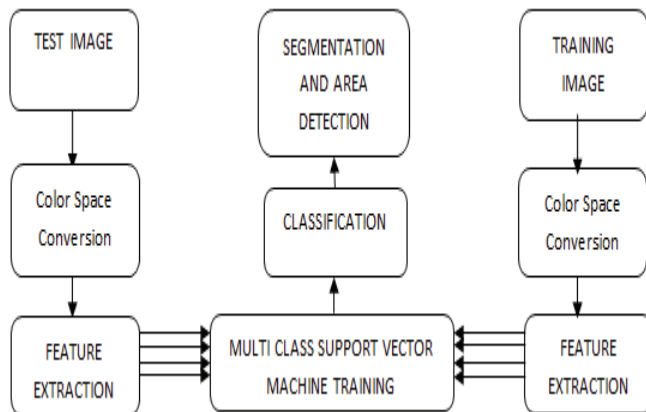


Fig. 1. Block Diagram of Proposed Method

From the figure 1, the test image is used to determine the type of cancer. The test image is converted from RGB to gray level color space. From the gray level region there are several built in features which are extracted as energy, auto correlation, gradient, contrast, standard deviation, correlation coefficient, skewness, homogeneity, cross correlation, mean, covariance, auto covariance, for both white and black regions. After the feature extraction, the extracted features are grouped in column wise manner.

In the same way, the identical features are extracted from the training samples. The training samples contain the predefined images for each type of cancer and also for normal. Each training samples are clustered by the

categorized type of cancer. In the MSVM classifier algorithm, the test samples are mapped with training samples and the probability value is set for the highest match obtained group of training samples. The probability values may differ for each type cancer images present in the database. Finally the feature samples are given to the classifier to detect and classify the type of the cancer. Once the classifier provides the classified type of cancer, the segmentation process begins to find the stages of abnormality and area of tumor [4] present in mm^2 .

III. PROPOSED METHOD

Proposed method contains four stages namely Image acquisition, Feature extraction, Multiclass Support Vector Machine and segmentation.

3.1 Image Acquisition

Acquisition of skin cancerous image is an important procedure in automated skin lesion [1] [10] analysis system. Because the acquired image will undergo the rest of the process in the system (such as feature extraction, MCSVM training, Segmentation and Area calculation) which may not be achievable or the obtained result will not be a reasonable one. Hence the acquired image should have healthy in resolution and clarity.

Usually an image can be acquired by any photographic equipment such as camera or mobile phones. Acquired image varies in resolution or pixel values by what type of equipment is used and for classification purposes the test is converted into 256x256 pixels. Normally the images are in the form of RGB color space but for feature extraction RGB color space is not suitable one. So that the color space conversion is done by conversion of RGB color space image into gray level image. These gray level converted images will be used for next process in system called feature extraction.

3.2 Feature Extraction

The accuracy and efficiency of the proposed system is based on extracted features from the skin cancer image. Texture and color features are the two main categorized features from extracted images. Autocorrelation, Contrast, maximum probability, sum of squares, sum average, sum variance etc., are the main features which are extracted with the help of Gray level co-occurrence (GLCM) matrix from the skin cancer image. With the help of GLCM there are some distinguished features that separate each class of skin cancers.

Eminent features of each class of skin cancerous are extracted with the help of feature extraction algorithm. Those extracted features are used for processing. Gray level co-occurrence matrix is the proposed feature

extraction technique. The acquired image converted from RGB to gray level, gray level is highly good for feature extraction using GLCM feature extraction technique. The pixel values of gray converted image are given to the GLCM matrix as an input, where the columns and rows of GLCM matrix is equal to the total number of pixels in the gray level image.

GLCM computes the frequency values for each pixels of the gray converted skin cancer image. The computed values from GLCM matrix are called as features or predictors. These extracted features are mapped with different angular positions by spatial relationship of different combination of pixels. Table 1 shows the list of features extracted.

Table 1. List of main features extracted

Features	Explanation
Energy	Measure the homogeneousness of the image.
Entropy	Measure of complexity of the image
Contrast	Measure of the local variations and texture of shadow depth
Homogeneity	Measure of closeness of the distribution of elements
Cross Correlation	Degree of joint probability occurrence of the specified pixel pairs.

3.3 Multiclass Support Vector Machine

MSVM is an extension of Support vector machine for solving the multiclass problems. Support vector machine is a very precise class of algorithms. With the help of kernels the SVM has been characterized the lack of local minima, the scarcity of the solution and the capability control obtained by acting on the margin. In case of Multi-class classification problems, the classification of more than one class is more complex due to the output of one class should be matched with more than one class. So that output of one class must be divided into M sub classes.

There are several methods available for solving the multi-classification problem. In this paper, it deals with the specific method called One-Against-All (OAA) classifier. The proposed OAA classifier is comparatively good from other classifier types such as Rest against one. 'M' binary SVM classifiers are made using One-Against-All classifier where the i^{th} SVM is trained such that the i^{th} class samples are labeled as positive samples and the remaining samples as negative samples. In the classification stage, the labeling of test samples are

obtained from all MSVM's is based on the maximum output among the 'M' classifier.

Let $S = \{(x_1, y_1), (x_2, y_2), \dots, (x_i, y_i)\}$ is a training sample set, Where x_i ($i=1 \dots N$) represent a vector of length n and y_j ($j=1 \dots M$) represents a class of sample. These sample points x_i and y_j are replaced with their mapping images $\phi(x_i)$ and $\phi(y_j)$ respectively. The method used to solve MCSVM classification problem is done by grouping of binary classification problems. In one against all method one constructs M classifiers, one for each class.

The M^{th} classifier constructs a hyper plane between class 'M' and remaining classes (M-1). For separating hyper plane we have to make decision surface in feature space. The weight and bias are introduced into it and given as follows.

$$\sum_{j=1}^i w_j \Phi(x) + b = 0 \quad (1)$$

The optimization problem is minimized by following expression

$$\Phi(w, f) = \frac{1}{2} \|w\|^2 + c \sum_{i=1}^l f_i \quad (2)$$

Where, f_i is a slack variable in order to relax the hard margin constraints and C is the regularization constant which implement the trade-off between margin of separation and error obtained in classification. The final classifier is of the form,

$$F(x) = \sum_{i=1}^l \alpha_i x_i^T x + b_M \quad (3)$$

3.4 Segmentation

An unrestrained growth of tissues in any part of the body is called as tumor and different types of tumors are available and each differs from its size and characteristics. Generally, melanoma images are taken by normal photograph and are needed to be well examined by physicians for diagnosing purposes. However the above method of detection resists the accuracy of determination of stage & size of tumor. In order to avoid the lack in accuracy, it is possible to propose a segmentation algorithm called K-means clustering algorithm.

K-Means algorithm is the unsupervised learning algorithm for clusters. According to the above said algorithm, the pixels are grouped or clustered to form an image. The number of clusters 'k' to be defined at initial process of k-means clustering algorithm. The center 'k' is chosen randomly. Once the k- cluster center is defined, a distance is calculated between pixels and cluster centers of each. A simple Euclidean function has been used for distance calculation. Using distance formula each pixels

are compared with all cluster centers. The shortest distance has been calculated using distance formula, based on the shortest distance among all the pixels are moved to a particular cluster.

Centroid is estimated once again and again each pixel is compared to all re-estimated centroids. Until the center converges, the above process is continued.

IV. RESULTS AND DISCUSSIONS

The simulation of Multi-class support vector machine to the test and training images provide classification of the cancer, which uses the built in features extracted from an images is acts as an input. In the resultant image, the region of cancer present is shown as a highlighted region with the help of segmentation and it provides the cancer type and the abnormality stage and also area of tumor present.

4.1 Simulation Results

There are five different skin lesions which are grouped as Solar Keratosis or actinic keratosis, Basal Cell Cancer, Nevocytic nevus, Squamous Cell Cancer, Seborrhoeic Verruca. Simulation had done for all type of melanoma and the corresponding value of tumor area is tabulated and shown in table 2.

Table 2. Type of melanoma and their values of tumor area

S. No.	Type of Melanoma	Value of Tumor Area (mm ²)
1	Solar Keratosis	65.1048
2	Basal Cell Carcinoma	66.7403
3	Nevocytic Nevus	59.4821
4	Squamous Cell Carcinoma	66.7231
5	Seborrhoeic Verrucas	62.8547

Simulation result is shown only for solar keratosis type of melanoma cancer. The figure 2 a) and 2 b) shows the simulation output of solar keratosis type of melanoma cancer. The classification and segmentation are shown using graphical user interface (GUI) window also area of tumor is calculated.

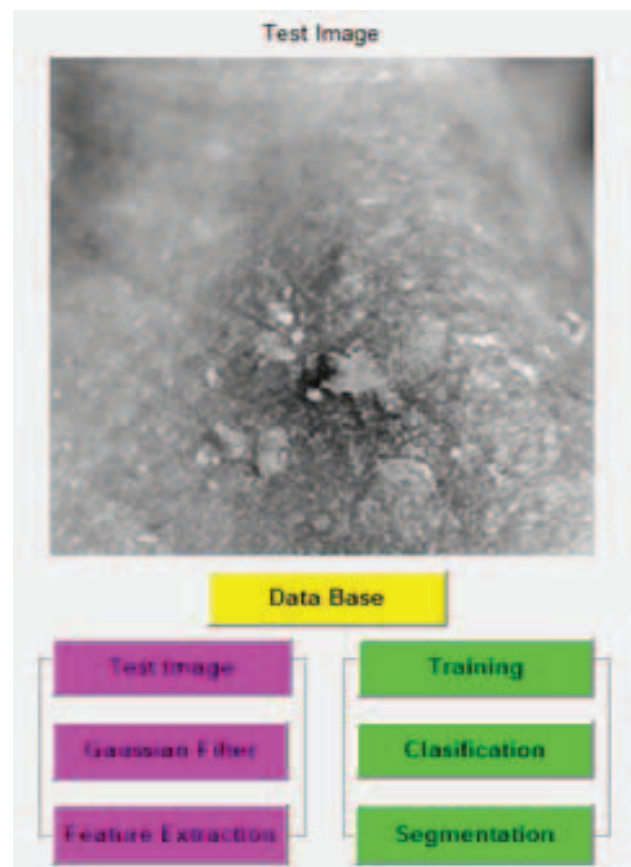


Fig. 2 a). Test image of Solar keratosis type skin cancer



Fig. 2 b). Other stages of Solar keratosis type skin cancer

V. CONCLUSION

The rate of patients with skin cancer will be a steadily increasing one due to the exposure of ultraviolet light from the sun into skin. It is hard to prevent and the effect can be accumulated. So that earlier detection plays an important role in diagnosis and treatment. Thus this paper has simulated a system called MCSVM classifier algorithm for early detection and classification. The simulation results have shown the superior performance and accuracy of One-Against-All MCSVM. The proposed algorithm is applied to all five types of melanoma skin cancers and the obtained results are segmented to find region and area of abnormality with higher accuracy.

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